Predictive Modeling: Utilizing Machine Learning to Forecast Diabetes Hospital Re-admissions

Author: Temitope Adereni
DataraFlow Internship Programme
Email: temitopeogundare2015@gmail.com
Supervised by: Winner Emeto (DataraFlow)

Abstract

Background: Hospital readmissions are a major and ongoing concern for healthcare systems worldwide, serving as a complex indicator of care quality while contributing billions in preventable costs each year. The likelihood of readmission among patients with diabetes mellitus is particularly high, reflecting the difficulties of managing multiple comorbid conditions and ensuring effective continuity of care.

Methods: This study proposes a systematic and reproducible framework for predicting 30-day hospital readmissions using the UCI Diabetes 130-US Hospitals dataset (Clore *et al*., 2014; https://doi.org/10.24432/C5230J), with additional external validation using a secondary hospital readmission dataset (https://www.kaggle.com/datasets/dubradave/hospital-readmissions). Logistic Regression and Random Forest classifiers are benchmarked using ROC-AUC, precision, recall, F1-score, and confusion matrices. Hyperparameters are explicitly reported for reproducibility. Class imbalance handling via SMOTE and class weighting is evaluated. SHAP (SHapley Additive exPlanations) values are applied for model interpretability.

Results: Logistic Regression demonstrated more stable discrimination performance (AUC = 0.624) than Random Forest (AUC = 0.609) on the UCI dataset. Utilization-related features significantly improved AUC from 0.529 to 0.629. SMOTE did not improve performance on the UCI dataset but yielded a modest recall improvement on the secondary dataset. SHAP analysis identified the number of prior inpatient visits, time in hospital, and emergency encounter frequency as the most influential predictors.

Conclusion: This study highlights the importance of feature selection, interpretable modelling, and cross-dataset validation in healthcare prediction. Utilization features consistently enhanced model discrimination, while SMOTE effects were dataset-dependent. The findings provide practical guidance for developing clinically meaningful readmission risk models.

Keywords: Hospital Readmission Prediction, Diabetes Mellitus, Machine Learning, Class Imbalance Handling, Explainable Artificial Intelligence (XAI), SHAP, Random Forest, Logistic Regression, Healthcare Data Analytics, Confusion Matrix, ROC-AUC

1. Introduction

Hospital readmissions are a persistent challenge to healthcare systems in the world. Often, readmissions are viewed as a marker of healthcare quality since frequent rehospitalization can reflect inadequate discharge planning, poor continuity of care, or unmanaged comorbidities [1]. Nevertheless, the financial burden is substantial. In the United States, the Centers for Medicare & Medicaid Services (CMS) estimated that unplanned readmissions account for over $17 billion in avoidable costs annually [2]. Internationally, readmission rates have been reported across a range of conditions, with significant variability between health systems, but consistently high costs and strain on resources [3].

Beyond economics, repeated hospitalizations are associated with worsened clinical outcomes, including increased mortality, hospital-acquired complications, and lower health-related quality of life [4,5]. Studies of older populations and patients with chronic disease confirm that readmission is linked to higher six-month and one-year mortality, as well as reduced patient well-being [6,7]. These findings underscore the urgent need for accurate and interpretable predictive tools that can identify patients at high risk of readmission before discharge.

Among chronic conditions, diabetes mellitus is one of the leading drivers of hospital utilization and readmissions [8, 9]. Its complex management involves glucose control, multiple medications and dietary adherence. This frequent monitoring creates opportunities for treatment gaps and complications that often result in returning to the hospital [9]. The UCI's Diabetes 130-US hospitals' dataset is a standard benchmark for investigating issues pertaining to readmission. The dataset covers over 100,000 patient encounters from 1999 to 2008 and includes demographic, diagnostic, and clinical variables [9]. Previous analyses of this dataset have demonstrated that patient age, length of stay, number of prior visits, and diagnosis categories are important predictors of readmission [10]. However, even with these insights, prediction accuracy remains limited and there is a continuous call for reproducible and interpretable methods that can be integrated into clinical and operational decision-making.

Existing research has provided a wide variety of approaches for hospital readmission prediction ranging from logistic regression models to advanced ensemble and deep learning techniques. However, several limitations persist. First, predictive models often exhibit limited generalizability when applied across different hospitals or patient populations. This means they struggle to be accurate or useful when applied to patients or healthcare settings other than the one they were originally trained on [11]. Second, many models underperform due to class imbalance, as readmissions occur less frequently than non-readmissions, leading to biased classifiers that favor the majority class [12]. Third, interpretability remains a concern: while clinicians value insights into why a patient is deemed at risk, many machine learning models are presented as "black boxes" without meaningful feature-level explanations [13]. Moreover, most prior studies emphasize algorithmic performance while neglecting the importance of clear preprocessing pipelines, standardized benchmarks, and clinically relevant visualization methods. There remains a gap in research that combines robust preprocessing, comparative benchmarking of interpretable models and visual exploration of risk factors.

This study aims to address these gaps through a structured and reproducible pipeline for predicting hospital readmissions using UCI Diabetes 130-US Hospitals for Years 1999-2008 dataset with a particular focus on diabetes-related admissions. In addition, external validation is incorporated to assess model generalizability across datasets. The contributions are fourfold:

1.1 Contributions

1. **Preprocessing pipeline**: Data cleaning, including handling of missing values, outlier detection, and standardized encoding of diagnosis codes.

2. **Comparative evaluation**: Benchmarking of Logistic Regression and Random Forest models with explicit hyperparameter reporting using ROC-AUC, PR-AUC, precision, recall, F1-score, calibration curves, and confusion matrices.

3. **Interpretability**: SHAP-based global and local feature importance analysis, identifying key clinical predictors of readmission risk.

4. **Empirical insights**: Practical insights for integrating predictive models into hospital decision support systems and real-time clinical workflows.

1.2 Scope and Limitations

This research is limited to the UCI Diabetes 130-US Hospitals dataset (1999-2008) and a secondary validation dataset. The study does not extend to unstructured EHR data such as clinical notes, imaging, or continuous monitoring signals. The focus is confined to patients with diabetes, and does not aim to develop a production-ready decision support system; rather, it emphasizes methodological benchmarking, interpretability, and visualization to inform future work.

2. Literature Review

The challenge of hospital readmission prediction has attracted substantial research attention over the past two decades, encompassing classical statistical methods, machine learning ensembles, deep learning architectures, and explainability frameworks.

2.1 Classical Predictive Models

In 2011, Kansagara et al. assessed twenty-six validated readmission risk models, finding that most exhibited poor discriminative power (c-statistics 0.55-0.65) due to limited representation of functional status and social determinants of health [11]. A systematic review by Artetxe et al. (2018) mapped 77 readmission prediction models, noting that logistic regression remained the dominant method (68% of studies), with AUC scores ranging from 0.54 to 0.92 and a critical absence of standardized benchmarks [12]. Strack et al. (2014) analysed 70,000 hospitalizations from the UCI Diabetes dataset, establishing it as the primary public benchmark for readmission research and demonstrating that HbA1c measurement was associated with reduced 30-day readmission [16].

2.2 Machine Learning and Deep Learning

Hu et al. (2022) developed an explainable machine learning model for in-hospital mortality prediction in septic ICU readmissions, achieving AUC = 0.81 with Random Forest alongside SHAP-based interpretability [13]. Hai et al. (2023) compared deep learning (LSTM) to traditional models on over 36,000 diabetic patients, with the LSTM achieving AUROC = 0.79 versus Random Forest at 0.72 [14]. Tang et al. (2023) introduced a multimodal spatiotemporal Graph Neural Network (MM-STGNN) for 30-day all-cause readmission, achieving AUROC = 0.724 by capturing complex time-dependent clinical variable relationships [15].

2.3 Explainability and Interpretability in Healthcare AI

Saranya et al. (2023) conducted a systematic review of explainable AI (XAI) across 91 healthcare studies, emphasizing the critical need for transparency in high-stakes clinical decision systems, and categorizing SHAP, LIME, and attention-based methods as the dominant interpretability approaches [17]. Their findings reinforce that clinical adoption of ML models requires not only accuracy but trustworthy explanations for individual predictions.

2.4 Recent Benchmarked Pipelines

Talwar et al. (2023) conducted the first meta-analysis directly comparing logistic regression and machine learning models for 30-day hospital readmission prediction in the US, synthesising nine studies identified from MEDLINE, PubMed, and Embase (2015–2019) [18]. Overall, ML modestly outperformed LR (AUC mean difference = 0.03; 95% CI: 0.01–0.05), though performance varied substantially across algorithm families: deep learning achieved the largest gain (MD = 0.06; 95% CI: 0.04–0.09), while tree-based methods such as Random Forest and boosting showed no statistically significant advantage over LR (MD = 0.02; 95% CI: −0.01–0.05). The certainty of evidence was rated very low due to observational study designs

and heterogeneity in clinical populations. Importantly, the authors highlighted that most reviewed models did not report precision, recall, or F1-score, and few addressed class imbalance — underscoring gaps that the present study explicitly addresses. Recent work (2021–2024) has increasingly focused on integrating real-time EHR signals, causal inference frameworks, and federated learning for privacy-preserving model development.

Searches were conducted across PubMed/PMC, arXiv, UCI/ML repos, Scopus/Google Scholar (2010–2025) using keywords such as “hospital readmission prediction”, “UCI diabetes readmission”, “random forest readmission”, “LSTM readmission EHR”, “readmission model calibration”, “SHAP LIME readmission.”

Inclusion criteria:

- Empirical models for hospital readmission (30-day or similar)
- Papers benchmarking ML methods or describing the UCI diabetes dataset
- Papers addressing evaluation/calibration/interpretability or imbalance handling

Exclusion Criteria

- Purely theoretical ML papers without clinical data
- Papers focused on other outcomes (mortality only)
- Studies older than 15 years (unless it is seminal)

Summary of Key Machine Learning and Statistical Models for Diabetes Readmission Prediction

Theme	Author	Dataset	Method	Strength	Weakness
Classical ML baselines (Logistic regression, decision trees, Random Forest)	Kansagara *et a*l. (2011)[11]	Literature review	Systematic review of statistical models	Comprehensive overview; sets clinical expectations for model performance	Older (pre-widespread ML) and limited in evaluating modern ensemble methods.
	Strack *et al*. (2014)[16]	UCI Diabetes 130-US Hospitals (1999–2008)	Logistic regression, baseline predictors	Large, public dataset widely used as a benchmark	Historic (1999–2008), missingness and coding idiosyncrasies require careful preprocessing
Deep learning & temporal models	Hai *et al*. (2023)[14]	Diabetes readmission dataset	Deep learning (LSTM, comparisons)	Shows potential gains from sequence models (LSTM) on longitudinal EHR	Gains are sometimes modest relative to well-tuned classical models, and interpretability suffers
	Tang *et al*. (2023)[15]	Large multimodal hospital dataset	Multimodal Spatiotemporal Graph Neural Network (MM-STGNN)	Models patient similarity and temporal patterns effectively	Complex, data-hungry, and harder to reproduce with tabular public datasets
Evaluation strategies/class imbalance/calibration	Artetxe *et al*. (2018)[12]	Literature review	Survey of ML (RF, SVM, ANN, etc.)	Broad benchmarking and attention to evaluation best practices	Rapidly evolving field (post-2018 DL advances not covered)
Interpretability & explainability	Hu *et al*. (2022)[13]	Hospital EHR data	Explainable ML (SHAP, LIME)	Shows how to move beyond black-box outputs to clinician-meaningful explanations.	Local explanations may vary across instances; needs careful presentation for stakeholders
	Saranya *et al*. (2023)[17]	Various healthcare datasets	Systematic review of XAI approaches	Strong guidance on combining methods and evaluating explanation reliability	Focuses heavily on imaging; applications to tabular readmission data need adaptation
Recent benchmarks	Talwar *et al*. 2023	Meta-analysis	Random-effects	First meta-analysis of ML vs. LR for	Heterogeneity in included studies

	[18]		meta-analysis; LR vs. ML comparison	readmission; subgroup analyses by algorithm type	complicates direct inference

3. Problem Statement

Hospital readmissions are a persistent problem for healthcare systems, contributing to high costs and poor patient outcomes. Diabetes mellitus is associated with frequent readmissions due to the complexity of managing comorbidities, medications, and follow-up care [1, 9]. Predictive modeling offers an opportunity to identify high-risk patients before discharge, but existing approaches often suffer from poor generalizability, difficulties with class imbalance, and insufficient interpretability [11, 12].

Given a patient encounter record $x \in X$ containing demographic, diagnostic, and utilization features, the task is to predict whether the patient will be readmitted within 30 days, represented as a binary label $y \in \{0, 1\}$. The objective is to design and benchmark models $f: X \rightarrow y$ that maximize predictive performance (AUC, precision, recall, F1-score) on imbalanced data, while providing interpretable feature-level explanations that are clinically actionable.

3.1 Research Questions and Hypotheses

RQ1: Can classical machine learning models such as Logistic Regression and Random Forest achieve reliable predictive accuracy for 30-day hospital readmissions in patients with diabetes?

RQ2: To what extent do utilization features (length of stay, prior admission history, emergency encounters) improve model performance compared with demographic-only predictors?

RQ3: Does addressing class imbalance (SMOTE, class weighting) significantly improve recall for predicting readmitted patients without compromising overall discrimination?

RQ4: How can SHAP-based visualization methods enhance the interpretability and clinical utility of predictive models for readmissions?

Hypotheses (Hs):

H1: Random Forest classifiers will outperform Logistic Regression in terms of discrimination (AUC, F1-score) on the diabetes readmission dataset.

H2: Incorporating utilization-related features (prior admissions, length of stay) will significantly improve prediction accuracy compared to models trained on demographic variables alone.

H3: Class imbalance correction methods will improve recall of readmitted cases while maintaining acceptable calibration.

4. Methodology

This study adopts an experimental research design to evaluate machine learning models for predicting 30-day hospital readmissions. The approach focuses on reproducible model development and controlled comparison of algorithms, feature sets, and class imbalance strategies. Rather than exploring complex deep learning architectures, this study prioritises interpretable baseline models — specifically Logistic Regression and Random Forest — to enable transparent evaluation and clinical relevance.

4.1 Dataset

Primary dataset: UCI Diabetes 130-US Hospitals dataset (1999-2008), sourced from the UCI Machine Learning Repository [9].

Citation: Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). Diabetes 130-US Hospitals for Years 1999-2008. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J

Size: 101,766 patient encounters from 130 US hospitals

Features: 50 variables including demographic, clinical, medication, and administrative data

Classes: Binary outcome; readmitted within 30 days (11.2%) vs. not readmitted (88.8%)

License: Open access for research and educational use

Known biases: Over-representation of certain age groups and ethnicities; data limited to U.S. hospitals (1999-2008); may not reflect current clinical practices

Secondary dataset: A secondary hospital readmission dataset was incorporated to evaluate model generalizability. The identical preprocessing and modeling pipeline was applied to both datasets.
Name: Hospital Readmissions Dataset (Kaggle)
Source: https://www.kaggle.com/datasets/dubradave/hospital-readmissions

4.2 Data Preprocessing

4.2.1 Record Exclusion and Column Removal

Administrative identifiers (encounter_id, patient_nbr) and high-missingness or low-signal variables were removed before modelling:

- encounter_id, patient_nbr - hospital system identifiers with no predictive value
- weight - 97.0% missing values; imputation would introduce unreliable estimates
- admission_type_id, discharge_disposition_id, admission_source_id - high-cardinality administrative codes
- payer_code - financial categorization not clinically relevant for readmission prediction

- medical_specialty - approximately 49.1% missing

Records after preprocessing: After removing high-missingness columns and applying inclusion/exclusion criteria (patients aged 18+, valid outcome labels), the final analytical dataset retained 101,766 encounters for the UCI dataset.

4.2.2 Missing Value Handling

Missing values encoded as '?' were converted to NaN before imputation:

- race: mode imputation (most frequent category = 'Caucasian'); 2.2% missing
- diag_1, diag_2, diag_3: converted to numeric; imputed with 0 (indicating no diagnosis code recorded)
- max_glu_serum, A1Cresult: filled with 'missing' category label (>70% absent; absence itself is clinically informative)

4.2.3 Encoding Categorical Variables

Method: One-Hot Encoding (pd.get_dummies, drop_first=True) was applied to all categorical variables during feature preparation. This avoids ordinal assumptions and produces binary indicator columns. Key encoded variables include: race, gender, age (binned), max_glu_serum, A1Cresult, and 23 medication variables (metformin, insulin, repaglinide, etc.).

4.2.4 Class Imbalance Assessment

The binary readmission target was highly imbalanced: 88.8% non-readmitted (Class 0) vs. 11.2% readmitted within 30 days (Class 1). This 7.9:1 imbalance ratio necessitates careful evaluation metric selection (F1-score, PR-AUC, Recall) and explicit imbalance correction experiments.

4.2.5 Normalization

Numerical features processed through the sklearn Pipeline ColumnTransformer were imputed with median values. StandardScaler was not applied to tree-based models (Random Forest is scale-invariant); for Logistic Regression, the built-in L2 regularization (C=1.0) mitigates the need for explicit scaling in this context. Future work should assess StandardScaler impact on LR calibration.

4.3 Model Specifications and Hyperparameters

Explicit hyperparameter settings are provided for reproducibility:

Model	Parameter	Value	Rationale
Logistic Regression	max_iter	1000	Ensure convergence on high-dimensional data

Logistic Regression	C (regularisation)	1.0	Default L2 regularization; prevents overfitting
Logistic Regression	solver	lbfgs	Suitable for multi-class and L2 penalty
Logistic Regression	class_weight	None (baseline) / 'balanced' (H3)	Explicit imbalance experiment
Random Forest	n_estimators	200	Sufficient trees for stable estimates
Random Forest	max_depth	12	Prevents overfitting to noise in sparse clinical data
Random Forest	min_samples_split	5	Regularizes leaf node splits
Random Forest	random_state	42	Fixed for reproducibility
SMOTE	random_state	42	Reproducible synthetic sample generation
All models	train/test split	80/20 stratified	Preserves class distribution; random_state=42

4.4 Evaluation Metrics

Model performance was evaluated with explicit attention to the class imbalance inherent in hospital readmission data:

- **ROC-AUC:** Primary discrimination metric; measures ability to rank patients by readmission risk across all thresholds
- **PR-AUC:** Precision-Recall area under curve; more informative than AUC for imbalanced datasets
- **Precision, Recall, F1-score:** Classification metrics at optimised decision threshold (F1-maximising threshold selected on validation set)
- **Confusion Matrix:** True positives, false positives, false negatives, true negatives; critical for clinical risk assessment where false negatives carry high consequence
- **Calibration curves:** Assess reliability of predicted probabilities for clinical decision support

4.5 Class Imbalance Strategies Evaluated

- **Baseline:** No correction applied

- **Class weighting:** class_weight='balanced' in Logistic Regression (inversely proportional to class frequencies)
- **SMOTE:** Synthetic Minority Oversampling Technique (k=5 neighbours, random_state=42) applied to training data only, preventing data leakage

4.6 Model Interpretability - SHAP

To explain individual and global predictions, SHAP (SHapley Additive exPlanations) was applied using the LinearExplainer for Logistic Regression. SHAP values quantify each feature's marginal contribution to the predicted readmission probability, averaged across coalition subsets. Both global summary plots (mean |SHAP|) and individual instance explanations were generated to support clinical transparency.

SHAP analysis was applied to the Logistic Regression model, as it demonstrated the most stable and consistent discrimination performance across datasets. While SHAP can also be applied to tree-based models using TreeExplainer, this study prioritised interpretability of the best-performing and most clinically transparent model. Future work may extend explainability analysis to Random Forest to compare feature attribution consistency across model classes.

5. Results

5.1 Model Benchmarking (H1)

Logistic Regression and Random Forest classifiers were evaluated on the UCI Diabetes dataset using an 80/20 stratified split. Logistic Regression achieved an AUC of 0.624 and F1-score of 0.028, whereas Random Forest achieved an AUC of 0.609 and F1-score of 0.012. Contrary to Hypothesis 1, the Random Forest did not outperform Logistic Regression. Both models exhibited low F1-scores, reflecting the severe class imbalance (11.2% minority class). H1 VERDICT: NOT SUPPORTED. Logistic Regression outperformed Random Forest on both AUC (0.624 vs. 0.609) and F1-score (0.028 vs. 0.012) on the UCI dataset, and replicated this advantage on the secondary dataset (AUC 0.644 vs. 0.605). The hypothesis that Random Forest would achieve superior discrimination is rejected.

Model	Dataset	AUC	PR-AUC	Precision	Recall	F1
Logistic Regression	UCI	0.624	0.152	0.119	0.495	0.028
Random Forest	UCI	0.609	0.130	0.088	0.412	0.012
Logistic Regression	Secondary	0.644	0.614	0.484	0.407	0.496
Random Forest	Secondary	0.605	0.571	0.476	0.512	0.537

Table 1: Cross-Dataset Model Performance Summary. Confusion matrix details for both models on both datasets are provided in Appendix B.

Cross-dataset validation on the secondary dataset confirmed the trend: Logistic Regression again achieved higher discrimination (AUC 0.644 vs. 0.605), while Random Forest demonstrated a higher F1-score (0.537 vs. 0.496), reflecting improved precision-recall balance on this dataset. The confusion matrix revealed that Random Forest detects more true positives at the cost of substantially more false positives.

5.2 Feature Set Comparison (H2)

Two model configurations were evaluated using Logistic Regression: (i) demographics-only (race, gender, age) and (ii) demographics combined with utilization features (time_in_hospital, number_inpatient, number_outpatient, number_emergency).

Model	Dataset	AUC	PR-AUC	Precision	Recall	F1
Demographics Only	UCI	0.529	0.120	0.119	0.765	0.205
Demo + Utilization	UCI	0.629	0.189	0.177	0.388	0.243
Demographics Only	Secondary	0.520	0.485	0.470	1.000	0.640

Demo + Utilization	Secondary	0.638	0.614	0.487	0.920	0.637

Table 2: Feature Set Comparison Across Datasets

On the UCI dataset, utilization features improved AUC from 0.529 to 0.629 (+0.100) and PR-AUC from 0.120 to 0.189. F1-score improved (0.205 → 0.243) despite a recall decrease, reflecting a shift toward a more conservative, precise classifier. On the secondary dataset, AUC improved from 0.520 to 0.638 (+0.118), confirming consistent signal from prior healthcare utilization across datasets. H2 VERDICT: SUPPORTED.

5.3 Class Imbalance Experiments (H3)

Strategy	Dataset	AUC	Precision	Recall	F1
Baseline	UCI	0.628	0.163	0.495	0.235
Class Weighting	UCI	0.628	0.162	0.495	0.235
SMOTE	UCI	0.606	0.180	0.423	0.235
Baseline	Secondary	0.636	0.642	0.407	0.493
Class Weighting	Secondary	0.635	0.640	0.407	0.493
SMOTE	Secondary	0.635	0.638	0.425	0.500

Table 3: Class Imbalance Strategy Comparison

On the UCI dataset, SMOTE oversampling decreased AUC from 0.628 to 0.606 and decreased recall from 0.495 to 0.423, while precision was 0.180. Confusion matrix analysis revealed a substantial increase in false positives without corresponding true positive gains. On the secondary dataset, SMOTE produced a modest recall improvement (0.407 → 0.425) while AUC remained effectively unchanged (0.636 → 0.635), and calibration remained acceptable. H3 VERDICT: PARTIALLY SUPPORTED (dataset-dependent).

5.4 ROC-AUC Summary

Model	Accuracy	F1	AUC	Calibration
Logistic Regression (UCI)	0.882	0.028	0.624	Good
Random Forest (UCI)	0.882	0.012	0.609	Good
LR + Utilization (UCI)	0.881	0.243	0.629	Good
LR + SMOTE (UCI)	0.870	0.235	0.606	Acceptable

Logistic Regression (Secondary)	0.594	0.484	0.644	Good
Random Forest (Secondary)	0.558	0.537	0.605	Acceptable

Table 4: Full Model Evaluation Summary

ROC-AUC figures from this study are broadly consistent with the meta-analytic literature: Talwar et al. (2023) [18] report that ML models achieve a mean AUC advantage of only 0.03 over LR (95% CI: 0.01-0.05) for 30-day readmission in US populations, and that tree-based methods such as Random Forest show no statistically significant AUC gain over LR, a finding directly replicated here (LR AUC 0.624 vs. RF AUC 0.609 on UCI).

Table 1 presents baseline model performance using the full feature set for model benchmarking (H1), whereas Table 4 includes feature-engineered variants (e.g., utilization features and SMOTE). Differences in F1-scores reflect these distinct experimental configurations rather than inconsistencies in model evaluation.

5.5 SHAP Model Interpretability

SHAP analysis applied to the best-performing Logistic Regression model identified the following key predictors of 30-day readmission:

- **number_inpatient:** Strongest predictor — prior inpatient visits are the most reliable indicator of future readmission risk, encoding chronic disease burden and healthcare dependency
- **time_in_hospital:** Longer stays indicate greater acute illness severity; positively associated with readmission
- **number_emergency:** Frequent emergency encounters signal unmanaged or deteriorating conditions
- **insulin:** Insulin treatment reflects glycaemic instability and higher disease complexity
- **age:** Older patients face compounding comorbidities and reduced functional reserve
- **number_outpatient:** Paradoxically, higher outpatient visits may reflect chronic disease monitoring (less readmission risk) or inadequately managed conditions (higher risk)

These findings align with established clinical literature on readmission drivers and provide actionable targets for care management interventions (e.g., enhanced discharge planning for patients with high number_inpatient scores).

6. Discussion

6.1 Model Performance

Across both datasets, Logistic Regression consistently achieved competitive or superior discrimination performance compared to Random Forest. On the UCI dataset, Logistic Regression outperformed Random Forest in AUC (0.624 vs. 0.609). This trend replicated on the secondary dataset (0.644 vs. 0.605). These findings suggest that the available administrative features contain limited nonlinear signals, and that noise and sparsity in clinical data reduce the advantage of more complex ensemble methods. This supports existing literature emphasising that simpler, interpretable models frequently match or exceed ensemble methods in administrative healthcare data [11, 12].

6.2 Importance of Utilization Features

The inclusion of utilization-related variables produced consistent improvements in discrimination across both datasets (AUC gains of +0.100 on UCI and +0.118 on secondary). These features likely encode latent disease severity, continuity of care, and patterns of healthcare engagement not reflected in demographic variables alone. While changes in Recall reflected shifts in the precision-recall trade-off, the overall improvement in ranking ability confirms that utilization features strengthen the model's ability to distinguish high-risk patients, consistent with clinical evidence that prior admissions are strong readmission predictors [8, 10].

6.3 Class Imbalance

The impact of class imbalance correction varied across datasets. On the UCI dataset, SMOTE oversampling reduced AUC and decreased recall, suggesting that synthetic samples introduced noise rather than useful minority class structure in this high-dimensional, heterogeneous clinical dataset. On the secondary dataset, SMOTE produced a modest recall improvement without compromising discrimination, indicating dataset-dependent behavior. These findings emphasise that imbalance correction strategies must be applied cautiously and validated empirically rather than assumed beneficial. Cost-sensitive learning and threshold optimisation may offer more stable alternatives.

6.4 Model Interpretability

SHAP analysis provided clinically meaningful insights into readmission drivers. The dominance of utilization features (number_inpatient, time_in_hospital, number_emergency) in SHAP

rankings corroborates the H2 findings and aligns with prior literature. Insulin treatment and age also emerged as influential, consistent with the known relationship between glycaemic instability, age-related comorbidities, and readmission risk [8, 16]. SHAP-based explanations support clinical trust and adoption by allowing clinicians to understand why a patient is classified as high-risk.

6.5 Limitations

This study has several limitations that should be considered when interpreting results:

- **Dataset scope:** Data limited to 130 U.S. hospitals (1999-2008); findings may not generalise to non-U.S. settings or contemporary clinical practices with different treatment protocols and EHR capabilities
- **EHR data bias:** The dataset over-represents certain age groups and ethnicities; models trained on this data may exhibit differential performance across demographic subgroups
- **No real-time deployment testing:** The study benchmarks models in a retrospective offline setting; prospective deployment in live clinical workflows introduces additional challenges (data drift, integration, clinician adoption)
- **Feature limitations:** Unstructured data (clinical notes, imaging, continuous monitoring signals) were not included; these may provide substantial additional predictive signal
- **Privacy considerations:** While the datasets are de-identified, linkage to real-time patient records for deployment would require rigorous privacy governance, institutional IRB review, and compliance with HIPAA/GDPR frameworks
- **Temporal confounding:** The dataset spans 1999-2008; treatment patterns, medication availability, and care protocols have changed substantially, potentially limiting the clinical applicability of model coefficients to current practice

7. Deployment Implications

The findings of this study have several practical implications for the integration of predictive readmission models into clinical and operational hospital workflows.

7.1 Integration into Hospital Decision Support Systems

Logistic Regression models with well-calibrated probabilities can be integrated into Electronic Health Record (EHR) platforms as real-time readmission risk scores. Because Logistic Regression produces interpretable coefficients and calibrated probability estimates, clinicians can understand both the predicted risk level and the contributing factors for individual patients. This supports discharge planning decisions, care coordination team activation, and post-discharge follow-up scheduling.

7.2 Real-Time Risk Alerts for Clinicians

A practical deployment architecture would compute readmission risk scores at discharge using routinely collected administrative and utilization data (number_inpatient, time_in_hospital, number_emergency, insulin use). SHAP-based explanations can be surfaced alongside risk scores in clinical dashboards, enabling nurses and discharge coordinators to identify the specific risk drivers for each patient and intervene accordingly — for example, arranging additional diabetes education for patients with poor glycaemic control, or scheduling earlier outpatient follow-ups for high-utilization patients.

7.3 Healthcare Cost Reduction

Identifying high-risk patients before discharge enables targeted preventive interventions: enhanced patient education, medication reconciliation, transitional care programs, and follow-up call protocols. The CMS Hospital Readmissions Reduction Program (HRRP) imposes financial penalties on hospitals with excess readmissions; a calibrated predictive model that improves identification of preventable readmissions could reduce penalty exposure and generate substantial cost savings. Estimates suggest even modest reductions in avoidable readmissions can save hundreds of millions of dollars annually across large hospital systems [2].

7.4 Ethical and Equity Considerations

Deployment of readmission risk models in clinical settings must account for potential algorithmic bias. Given the dataset's known demographic imbalances, models should be evaluated for differential performance across racial and ethnic subgroups before deployment. Fairness audits, disparate impact assessments, and regular model retraining are essential components of

responsible clinical AI deployment. Clinicians must be trained to use model outputs as decision support, not as autonomous decision-makers, to maintain human oversight and accountability.

This study used publicly available, de-identified datasets and did not involve direct patient contact or access to identifiable personal health information. As such, institutional ethical approval and informed consent were not required.

8. Conclusion

This study evaluated machine learning models for predicting 30-day hospital readmissions using demographic and utilization features across two independent datasets. Logistic Regression consistently achieved comparable or better discrimination performance than Random Forest, indicating that simpler, interpretable models can generalise effectively in administrative healthcare data contexts.

The inclusion of utilization-related variables (number_inpatient, time_in_hospital, number_emergency) significantly improved predictive performance across both datasets, highlighting the importance of prior healthcare interactions as indicators of readmission risk. SHAP analysis corroborated these findings, identifying utilization features as the dominant clinical drivers.

The impact of class imbalance correction was dataset-dependent. While SMOTE degraded performance on the UCI dataset, it produced a modest recall improvement on the secondary dataset. These findings emphasize that imbalance handling must be empirically evaluated rather than applied by default.

This study provides a reproducible, preprint-ready framework for readmission prediction that can serve as a foundation for clinical deployment, model refinement, and future research incorporating richer data sources such as temporal EHR trajectories and unstructured clinical notes.

8.1 Future Work

Future research directions include:

- Richer feature representations: temporal patient trajectories, medication patterns, and comorbidity embeddings to better capture disease progression
- Advanced imbalance strategies: cost-sensitive learning, focal loss, and threshold optimisation for more stable minority class sensitivity
- Expanding validation to larger, multi-institutional, multi-country datasets to assess model transportability
- Incorporating explainable AI tools (SHAP, LIME) in prospective clinical pilot studies to evaluate clinician acceptance and decision quality
- Federated learning approaches to enable model training across hospital systems without sharing patient-level data

- Fairness-aware model development to ensure equitable performance across demographic subgroups

References

1. J. M. McIlvennan, L. Eapen, and L. Allen, “Hospital Readmissions Reduction Program,” *Circulation*, vol. 131, no. 20, pp. 1796–1803, 2015.

2. S. R. Jencks, M. V. Williams, and E. A. Coleman, “Rehospitalizations among patients in the Medicare fee-for-service program,” *N. Engl. J. Med.*, vol. 360, no. 14, pp. 1418–1428, 2009.

3. F. Foroutan et al., “Global comparison of readmission rates for patients with heart failure,” *J. Am. Coll. Cardiol.*, vol. 82, no. 5, pp. 430-444, 2023.

4. K. Dharmarajan, Y. Wang, Z. Lin, S. T. Normand, J. S. Ross, L. I. Horwitz, N. R. Desai, L. G. Suter, E. E. Drye, S. M. Bernheim, and H. M. Krumholz, “Association of changing hospital readmission rates with mortality rates after hospital discharge,” *JAMA*, vol. 318, no. 3, pp. 270–278, Jul. 2017, doi: 10.1001/jama.2017.8444.

5. J. S. Rachoin, K. Hunter, J. Varallo, and E. Cerceo, "Impact of time from discharge to readmission on outcomes: an observational study from the US National Readmission Database," *BMJ Open*, vol. 14, no. 8, pp. e085466, Aug. 2024, doi: 10.1136/bmjopen-2024-085466.

6. J. Andreasen, R. J. J. Gobbens, H. H. Eriksen, and K. Overvad, "Health-related quality of life at hospital discharge as a predictor for 6-month unplanned readmission and all-cause mortality of acutely admitted older medical patients," *Qual. Life Res.*, vol. 28, no. 11, pp. 3015–3024, Nov. 2019, doi: 10.1007/s11136-019-02259-w.

7. T. Leosdottir et al., “Self-reported health-related quality of life predicts five-year mortality and hospital readmission in patients with ischaemic heart disease,” *Eur. J. Prev. Cardiol.*, vol. 21, no. 3, pp. 389–396, 2014.

8. D. J. Rubin and A. A. Shah, "Predicting and Preventing Acute Care Re-Utilization by Patients with Diabetes," *Curr. Diab. Rep.*, vol. 21, no. 9, p. 34, 2021. doi: 10.1007/s11892-021-01402-7.

9. J. Clore, K. Cios, J. DeShazo, and B. Strack, “Diabetes 130-US Hospitals for Years 1999-2008,” *UCI Machine Learning Repository*, 2014. Accessed: Oct. 10, 2025. [Online]. Available: https://doi.org/10.24432/C5230J.

10. O. G. Emi-Johnson and K. J. Nkrumah, "Predicting 30-day hospital readmission in patients with diabetes using machine learning on electronic health record data," *Cureus*, vol. 17, no. 4, 2025.

11. D. Kansagara, H. Englander, A. Salanitro, D. Kagen, C. Theobald, M. Freeman, and S. Kripalani, “Risk prediction models for hospital readmission: a systematic review,” *JAMA*, vol. 306, no. 15, pp. 1688–1698, 2011.

12. A. Artetxe, A. Beristain, and M. Graña, “Predictive models for hospital readmission risk: A systematic review of methods,” *Comput. Methods Programs Biomed*., vol. 164, pp. 49–64, Oct. 2018, doi: 10.1016/j.cmpb.2018.06.006.

13. C. Hu, L. Li, Y. Li, F. Wang, B. Hu, and Z. Peng, "Explainable Machine-Learning Model for Prediction of In-Hospital Mortality in Septic Patients Requiring Intensive Care Unit Readmission," *Infect. Dis. Ther*., vol. 11, no. 4, pp. 1695–1713, Aug. 2022, doi: 10.1007/s40121-022-00671-3.

14. A. Hai et al., "Deep Learning vs Traditional Models for Predicting Hospital Readmission among Patients with Diabetes," in *AMIA Annu. Symp. Proc*., 2022, pp. 512–521.

15. S. Tang *et al.*, "Predicting 30-day all-cause hospital readmission using multimodal spatiotemporal graph neural networks," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 4, pp. 2071–2082, Apr. 2023.

16. B. Strack et al., ‘Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records’, *BioMed Research International*, vol. 2014, pp. 1–11, 2014, doi: 10.1155/2014/781670.

17. A. Saranya and R. Subhashini, "A systematic review of explainable artificial intelligence models and applications: Recent developments and future trends," *Decis. Anal. J.*, vol. 7, p. 100230, 2023, doi: 10.1016/j.dajour.2023.100230.

18. A. Talwar et al., "Performance of advanced machine learning algorithms over logistic regression in predicting hospital readmissions: A meta-analysis," *Healthcare Analytics*, vol. 4, p. 100218, 2023. https://doi.org/10.1016/j.health.2023.100218 (PMCID: PMC10474076)

Appendix

Appendix A - Repository

Full code and notebooks available at:

https://github.com/temitopeogundare2015-blip/DiabetesGuard-AI

Appendix B - Confusion Matrix Details

Confusion matrix values for the key model comparisons on the UCI dataset:

Model	True Negatives	False Positives	False Negatives	True Positives
Logistic Regression (baseline)	17,819	69	2,266	199
Random Forest	17,871	17	2,312	153
LR + Utilization Features	17,680	208	2,356	109
LR + SMOTE	16,950	938	2,450	15

Note: High false negative counts across all models reflect the severe class imbalance (11.2% minority class). The SMOTE model dramatically increases false positives without proportional true positive gains, confirming noise introduction.